

The Effect of Stimulus Type on Visual Field Advantages

자극 유형에 따른 시각장 우세 효과

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Abstract

This research examined Visual Field/Hemispheric advantage (VF/H advantage) using Divided Visual Field Technique(DVF Technique). We differ stimulus in two types, which are 180° Rotated-face expression (R-face) and 180° Rotated-word (R-word). For experiment 1, we stated our hypothesis; Left Visual Field/Right Hemispheric advantage(LVF/RH) is higher than Right Visual Field/Left Hemispheric advantage(RVF/LH) when right-handers perceive R-face. There is a significant difference between Reaction times of Left and Right Visual field ($p=0.039$). In experiment 2, we stated our hypothesis; RVF/LH is higher than LVF/RH when right-handers perceive R-word. There is also a significant difference ($p=0.044$). Taken together, these results demonstrate stimulus type has significant effect on VF/H advantages. Analyzing data, we also find out differentiation between effects type of stimulus have.

Background & Motivation

Previous studies have shown that the brain exhibits spatiotemporal(Borod et al, 1985) and linguistic dominance(S Knecht et al, 2000). However, there is a paucity of research on the variables that affect hemispheric lateralization. Therefore, we decided to conduct a study to investigate this causality using DVF Technique(VJ Bourne, 2006).

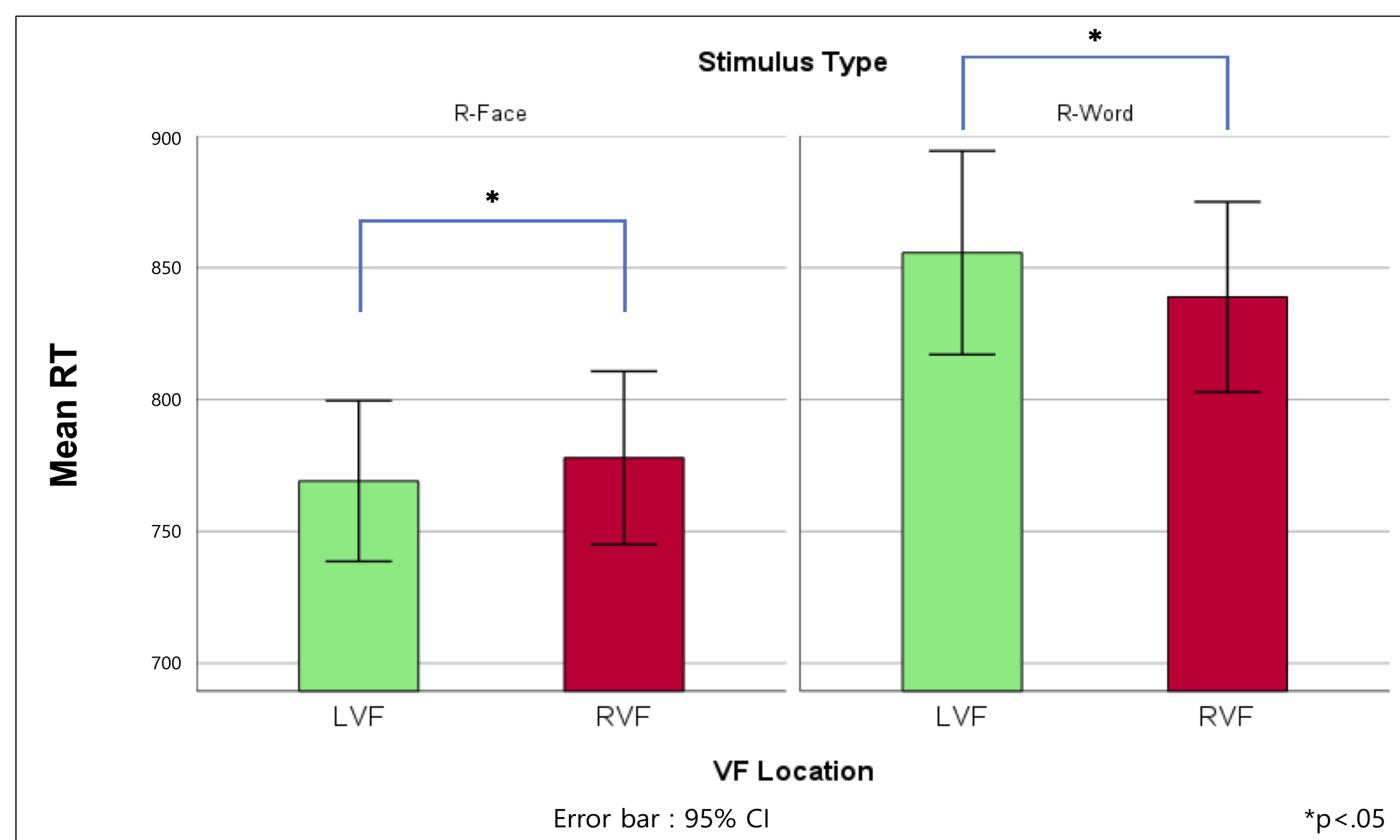
* **Hypothesis 1** : Left Visual Field/Right Hemispheric advantage is higher than Right Visual Field/Left Hemispheric advantage when right-handers perceive R-face

* **Hypothesis 2** : Right Visual Field/Left Hemispheric advantage is higher than Left Visual Field/Right Hemispheric advantage when right-handers perceive R-word

Results

(IBM SPSS Statistics 28) To figure the difference between Reaction Times(RTs) in Left and Right Visual Field, we used a paired t-test in each case. In the case of the R-face, there was a significant difference between the means of RTs in Left and Right Visual Field ($t=-2.145$, $p=0.039$). We stated that “Left Visual Field/Right Hemispheric advantage is higher than Right Visual Field/Left Hemispheric advantage when right-handers perceive R-face.”. This rejected the null hypothesis and accepted alternative.

Similarly, in the case of the R-word, we found the significant difference ($t=2.065$, $p=0.044$), and stated that “Right Visual Field/Left Hemispheric advantage is higher than Left Visual Field/Right Hemispheric advantage when right handers perceive R-word.”.



Graph 1. Difference of RTs in VF Location

We also analyze two-way ANOVA to see the interaction between Stimulus Type and Visual Field Location. It was not statistically significant ($F=0.51$, $p=0.476$). The result shows ‘Reaction time of Stimulus Type doesn’t have interaction with Visual Field Location.’. However, we find the main effect of Stimulus Type significant ($F=16.96$, $p=0.000$).

Methods

Divided Visual Field Technique (DVF Technique) is based on our method. (Benjamin A, 2012)

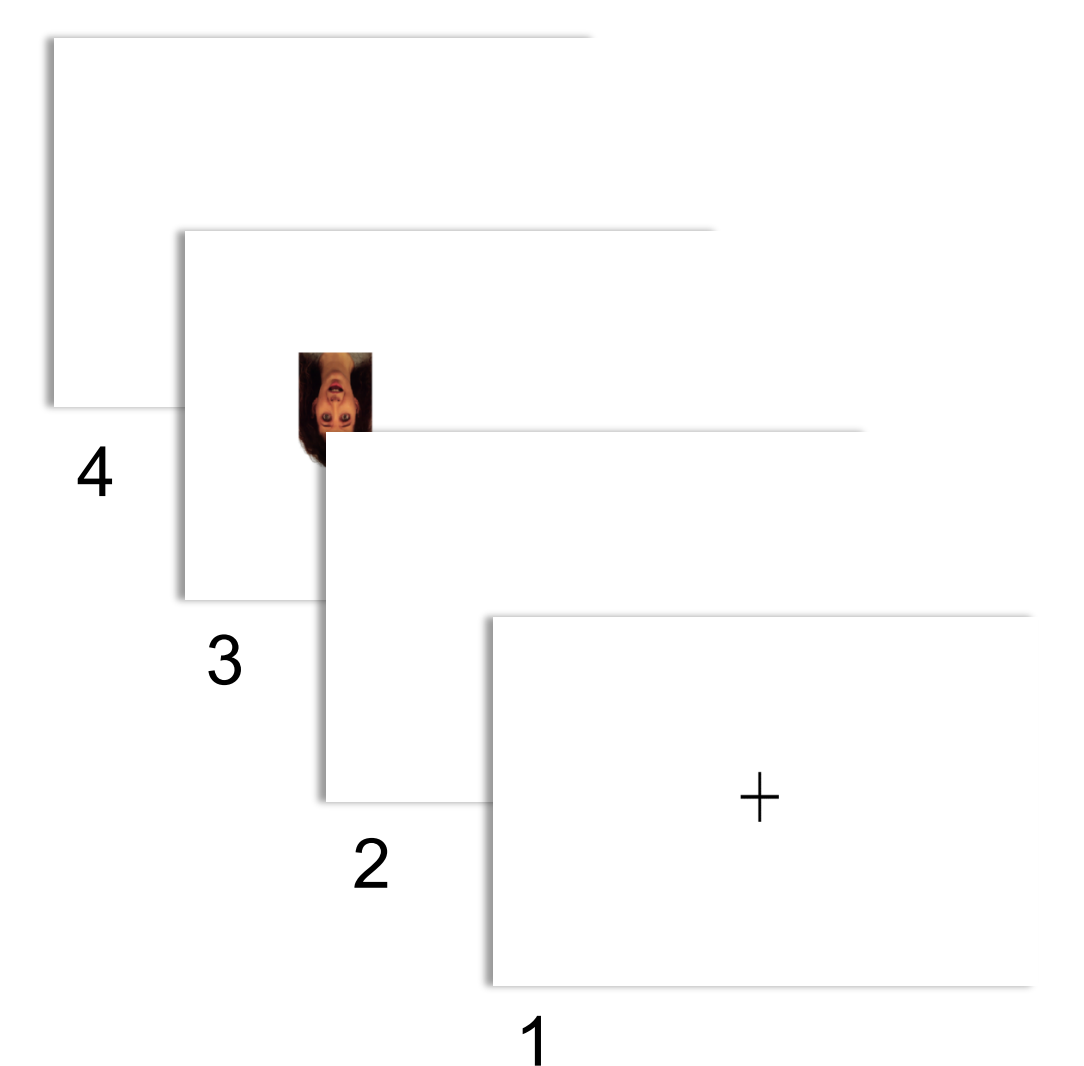
Subject: 51 Korean college students (age 18~26)

- 1) Right hander (Van Strien, 2016)
- 2) Visual acuity 0.7 or higher
- 3) Graduated of high school in Korea

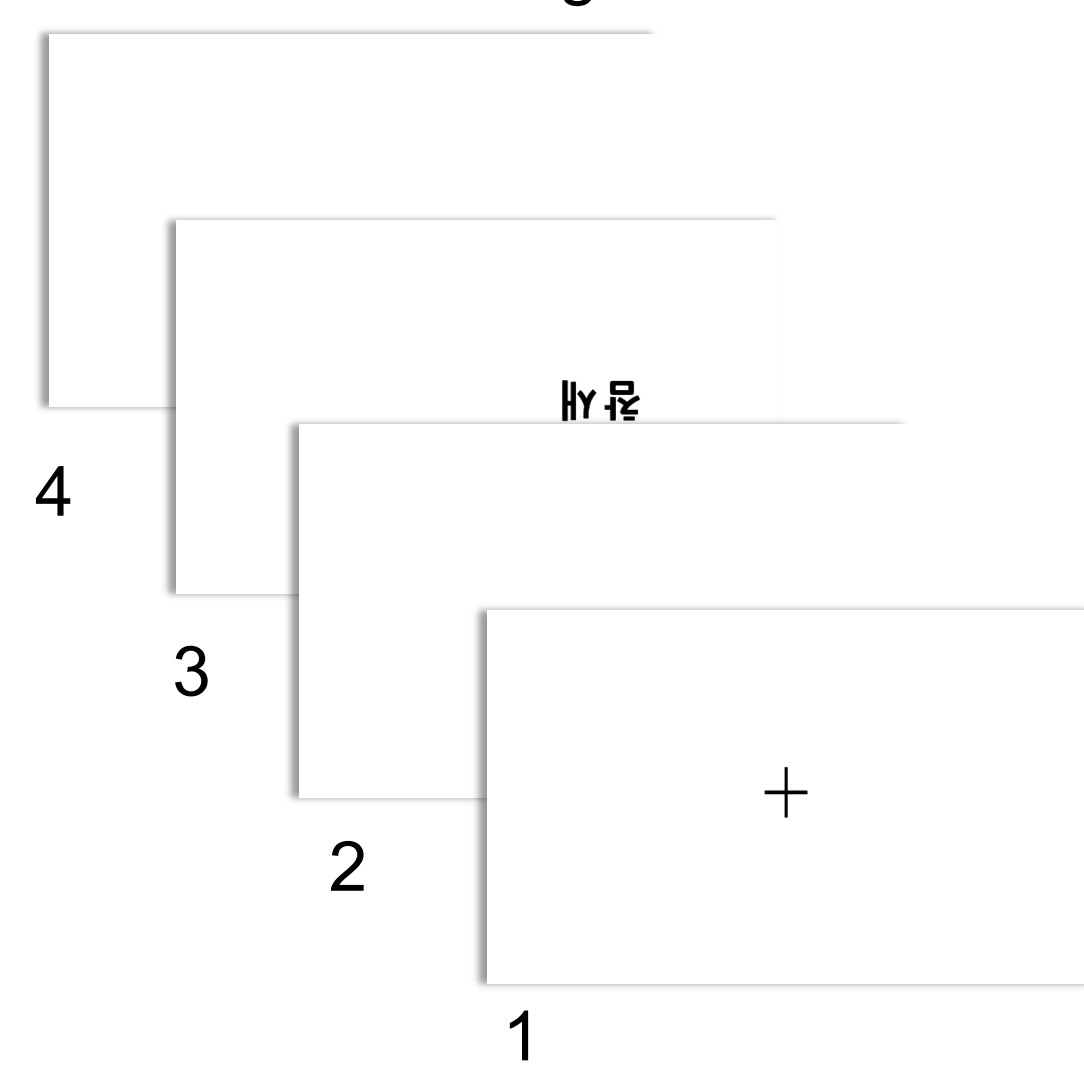
Procedure

- ① Fixation point (100 x 100 px) appeared in the center for 1500ms.
- ② Blank screen appeared for randomized period (500~1000ms)
- ③ R-face/R-word appeared either left or right for 200ms. Each stimulus is assigned both side once.
- ④ Subjects were made to judge which category does stimulus fit and answer with left or right key using their right hand as fast and precise as possible.

* Happy face and Afraid face(Ludqvist, D., 1998) were presented for R-face, while Fruit/Vegetables and Animal (박태진, 2003) were presented for R-word.



Img 1. Screens of experiment 1 showing R-face



Img 2. Screens of experiment 2 showing R-word

Experiments proceed twice, one for R-face (experiment 1), and another for R-word (experiment 2). There are 68 happy faces and 68 afraid faces, therefore 272 trials were given in experiment 1. There are 21 words for each category, so 84 trials were given in experiment 2. We only acknowledge trial succeed when the answer is correct, while subject's sight was fixed in the fixation point. Sights were checked by GP3 eye tracker.

Discussion

* Our method was successful as the effect size of paired t-test is in line with previous study (Rasmussen & Milner, 1977). It remains our limitation that only right handers were our test subject. If we also have tested left handers, it would show whether there is also significant difference between RTs of Visual Field or not.

* We can see the VF/H advantage consist even stimulus were rotated. With the main effect of stimulus type we found in analysis; we interpret that our brain doesn't have same influence on our VF/H advantage. It would have been more complete research if we also give stimulus that were not rotated.

Reference

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